Eby Friedman: Advancing high-complexity, high-performance integrated circuit design

Friedman was recognized for his pioneering work in high-performance integrated circuits. Friedman's work spans topics such as clock synchronization, power delivery and management, interconnect models, three-dimensional (3-D) integration, and emerging technologies such as superconductive circuits and magnetic devices. His research is used in a variety of common applications, such as microprocessors, wireless cell phones, cloud computing centers, and electronic design automation (EDA) tools used throughout industry.

As an inventor, Friedman holds 25 US patents and four foreign patents, which cover topics ranging from circuits for high-performance microprocessors and superconductive systems to methods for directional biasing of magnetic devices in high-density magnetic memory. His research has been integrated into a variety of commercial products across a diverse group of industries such as ink jet printers (Xerox), electronic design automation algorithms (Synopsys), digital cameras (Eastman Kodak), power management circuits (Qualcomm), and synchronous circuits (Intel) for microprocessors. His patents have been licensed by the primary EDA companies.

Friedman has been a member of several technical advisory boards and has founded multiple startup companies in the areas of electronic design automation and quantum computing. Several of these startups commercialized design tools for clock network synthesis to automate the on-chip synchronization process, necessary in almost all modern integrated circuits.

Friedman worked at Hughes Aircraft Company for 12 years, where he participated in the development of a large variety of integrated circuits and systems for US military and commercial applications. Friedman joined the University of Rochester in 1991 where he directs the High-Performance VLSI/IC Design and Analysis Laboratory. Friedman also serves as a visiting professor at the Technion—Israel Institute of Technology, where he directs the Technion Advanced Circuits Research Center and is a member of the Technion Board of Governors. He previously led the Center for Electronic Imaging Systems at Rochester for nine years. The Center is now known as the **Center for Emerging and Innovative Sciences (https://www.ceis.rochester.edu/)**.

A prolific author and scholar, Friedman has authored more than 600 peer-reviewed publications and 17 textbooks. His books cover a broad range of topics, including high performance integrated circuits, 3-D integrated circuits, quantum integrated circuits, and on-chip power delivery and management. Friedman is a Fellow of the IEEE and past Editor-in-Chief of the *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* and the *Microelectronics Journal*.

Friedman is a Fulbright scholar and has been a recipient of many awards such as a University of Rochester College of Engineering Teaching Excellence Award, the University of Rochester Graduate Teaching Award, the University of Rochester Hajim Lifetime Achievement Award, the IEEE Circuits and Systems Charles A. Desoer Technical Achievement Award, the IEEE Circuits and Systems Mac Van Valkenburg Award, and is an inaugural member of the University of California, Irvine Engineering Hall of Fame.